

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{144} c_\gamma \frac{1}{m_\Phi^2} \left(m_\Phi^2 y_d^{i1i2} \left(-96 g_3^2 + 27 g_2^2 \left(3 + 2 c_\gamma^2 + 2 s_\gamma^2 \right) + g_1^2 \left(31 + 18 c_\gamma^2 + 18 s_\gamma^2 \right) \right) + \right. \right. \\
& \quad 54 \overline{y_d}^{pr} \left(m_\Phi^2 y_d^{pi2} y_d^{i1r} \left(1 + s_\gamma^2 \right) - 4 C_{H^2} s_\gamma^2 y_d^{pr} y_d^{i1i2} \right) + \\
& \quad \left. 18 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} \left(m_\Phi^2 \left(1 + c_\gamma^2 \right) - 2 C_{H^2} s_\gamma^2 \right) \right) + \\
& \quad \frac{1}{2} \sum_p s_{2\gamma} s_\gamma g_1^2 \frac{1}{m_\Phi^4} y_d^{i1i2} \left(C_{H^2} - m_\Phi^2 \right) LF_{1,0} \left[m_d^{p\cdot} \right] + \frac{3}{2} s_{2\gamma} s_\gamma \frac{1}{m_\Phi^4} \overline{y_d}^{pr} y_d^{pr} y_d^{i1i2} \\
& \quad \left(-C_{H^2} + m_\Phi^2 \right) LF_{1,0} \left[m_d^{r\cdot} \right] + \frac{1}{2} \sum_p s_{2\gamma} s_\gamma g_1^2 \frac{1}{m_\Phi^4} y_d^{i1i2} \left(C_{H^2} - m_\Phi^2 \right) LF_{1,0} \left[m_e^{p\cdot} \right] + \\
& \quad \frac{1}{2} s_{2\gamma} s_\gamma \frac{1}{m_\Phi^4} y_d^{i1i2} \overline{y_e}^{pr} y_e^{pr} \left(-C_{H^2} + m_\Phi^2 \right) LF_{1,0} \left[m_e^{r\cdot} \right] + \\
& \quad \frac{1}{2} s_{2\gamma} s_\gamma \frac{1}{m_\Phi^4} y_d^{i1i2} \left(-C_{H^2} + m_\Phi^2 \right) \left(\overline{y_e}^{pr} y_e^{pr} + \sum_p g_1^2 \right) LF_{1,0} \left[m_l^{p\cdot} \right] - \\
& \quad \frac{1}{2} s_{2\gamma} s_\gamma \frac{1}{m_\Phi^4} y_d^{i1i2} \left(-C_{H^2} + m_\Phi^2 \right) \left(-3 \overline{y_d}^{pr} y_d^{pr} + 3 \overline{y_u}^{pr} y_u^{pr} + \sum_p g_1^2 \right) LF_{1,0} \left[m_q^{p\cdot} \right] + \\
& \quad \sum_p s_{2\gamma} s_\gamma g_1^2 \frac{1}{m_\Phi^4} y_d^{i1i2} \left(-C_{H^2} + m_\Phi^2 \right) LF_{1,0} \left[m_u^{p\cdot} \right] + \\
& \quad \frac{3}{2} s_{2\gamma} s_\gamma \frac{1}{m_\Phi^4} y_d^{i1i2} \overline{y_u}^{pr} y_u^{pr} \left(C_{H^2} - m_\Phi^2 \right) LF_{1,0} \left[m_u^{r\cdot} \right] + \\
& \quad \frac{3}{8} s_{4\gamma} s_\gamma \frac{1}{m_\Phi^4} y_d^{i1i2} \left(g_1^2 + g_2^2 \right) \left(C_{H^2} - m_\Phi^2 \right) LF_{1,0} \left[m_\Phi \right] - \\
& \quad \frac{1}{4} c_\gamma y_d^{pi2} \left(3 s_\gamma^2 \overline{y_d}^{pr} y_d^{i1r} + \overline{y_u}^{pr} y_u^{i1r} \left(c_\gamma^2 + 4 s_\gamma^2 \right) \right) LF_{1,1} \left[m_\Phi \right] - \\
& \quad \frac{1}{2} C_{H^2} c_\gamma s_\gamma^2 y_d^{pi2} \overline{y_u}^{pr} y_u^{i1r} LF_{1,2} \left[m_\Phi \right] - \\
& \quad \frac{1}{9} c_\gamma g_1^2 y_d^{i1i2} LF_{1,1,0} \left[m_1, m_d^{i2} \right] + \frac{1}{18} c_\gamma g_1^2 y_d^{i1i2} LF_{2,1,-1} \left[m_1, m_d^{i2} \right] - \\
& \quad \frac{1}{36} c_\gamma g_1^2 y_d^{i1i2} LF_{1,1,0} \left[m_1, m_q^{i1} \right] + \frac{1}{72} c_\gamma g_1^2 y_d^{i1i2} LF_{2,1,-1} \left[m_1, m_q^{i1} \right] + \\
& \quad m_1 s_\gamma \tilde{\mu} g_1^2 \frac{1}{m_\Phi^4} y_d^{i1i2} \left(C_{H^2} - m_\Phi^2 \right) \left(-c_\gamma^2 + s_\gamma^2 \right) LF_{1,1,0} \left[m_1, \tilde{\mu} \right] - \\
& \quad \frac{3}{4} c_\gamma g_2^2 y_d^{i1i2} LF_{1,1,0} \left[m_2, m_q^{i1} \right] + \frac{3}{8} c_\gamma g_2^2 y_d^{i1i2} LF_{2,1,-1} \left[m_2, m_q^{i1} \right] + \\
& \quad 3 m_2 s_\gamma \tilde{\mu} g_2^2 \frac{1}{m_\Phi^4} y_d^{i1i2} \left(C_{H^2} - m_\Phi^2 \right) \left(-c_\gamma^2 + s_\gamma^2 \right) LF_{1,1,0} \left[m_2, \tilde{\mu} \right] - \\
& \quad \frac{4}{3} c_\gamma g_3^2 y_d^{i1i2} LF_{1,1,0} \left[m_3, m_d^{i2} \right] + \frac{2}{3} c_\gamma g_3^2 y_d^{i1i2} LF_{2,1,-1} \left[m_3, m_d^{i2} \right] - \\
& \quad \frac{4}{3} c_\gamma g_3^2 y_d^{i1i2} LF_{1,1,0} \left[m_3, m_q^{i1} \right] + \frac{2}{3} c_\gamma g_3^2 y_d^{i1i2} LF_{2,1,-1} \left[m_3, m_q^{i1} \right] + \\
& \quad 3 s_\gamma \frac{1}{m_\Phi^4} y_d^{i1i2} \left(C_{H^2} - m_\Phi^2 \right) \left(s_\gamma \overline{a_d}^{pr} + \tilde{\mu} c_\gamma \overline{y_d}^{pr} \right) \left(-c_\gamma a_d^{pr} + s_\gamma \tilde{\mu} y_d^{pr} \right) LF_{1,1,0} \left[m_d^{r\cdot}, m_q^{p\cdot} \right] - \\
& \quad \frac{1}{2} c_\gamma \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{1,1,0} \left[m_d^{r\cdot}, \tilde{\mu} \right] + \\
& \quad s_\gamma \frac{1}{m_\Phi^4} y_d^{i1i2} \left(C_{H^2} - m_\Phi^2 \right) \left(s_\gamma \overline{a_e}^{pr} + \tilde{\mu} c_\gamma \overline{y_e}^{pr} \right) \left(-c_\gamma a_e^{pr} + s_\gamma \tilde{\mu} y_e^{pr} \right) LF_{1,1,0} \left[m_e^{r\cdot}, m_l^{p\cdot} \right] + \\
& \quad \frac{1}{2} \frac{1}{m_\Phi^2} y_d^{i1i2} \left(\overline{a_e}^{pr} \left(m_\Phi^2 c_\gamma^2 + 2 C_{H^2} s_\gamma^2 \right) + s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \left(2 C_{H^2} - m_\Phi^2 \right) \right) \\
& \quad \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr} \right) LF_{2,1,0} \left[m_l^{p\cdot}, m_e^{r\cdot} \right] - \frac{1}{2} \frac{1}{m_\Phi^2} y_d^{i1i2} \\
& \quad \left(\overline{a_e}^{pr} \left(m_\Phi^2 c_\gamma^2 + 2 C_{H^2} s_\gamma^2 \right) + s_\gamma \tilde{\mu} c_\gamma \overline{y_e}^{pr} \left(2 C_{H^2} - m_\Phi^2 \right) \right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr} \right) LF_{3,1,-1} \left[m_l^{p\cdot}, m_e^{r\cdot} \right] + \\
& \quad C_{H^2} c_\gamma y_d^{i1i2} \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr} \right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr} \right) LF_{3,1,0} \left[m_l^{p\cdot}, m_e^{r\cdot} \right] - \\
& \quad 3 C_{H^2} c_\gamma y_d^{i1i2} \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr} \right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr} \right) LF_{4,1,-1} \left[m_l^{p\cdot}, m_e^{r\cdot} \right] + \\
& \quad 2 C_{H^2} c_\gamma y_d^{i1i2} \left(c_\gamma \overline{a_e}^{pr} - s_\gamma \tilde{\mu} \overline{y_e}^{pr} \right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr} \right) LF_{5,1,-2} \left[m_l^{p\cdot}, m_e^{r\cdot} \right] + \\
& \quad \frac{3}{2} \frac{1}{m_\Phi^2} y_d^{i1i2} \left(\overline{a_d}^{pr} \left(m_\Phi^2 c_\gamma^2 + 2 C_{H^2} s_\gamma^2 \right) + s_\gamma \tilde{\mu} c_\gamma \overline{y_d}^{pr} \left(2 C_{H^2} - m_\Phi^2 \right) \right) \\
& \quad \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr} \right) LF_{2,1,0} \left[m_q^{p\cdot}, m_d^{r\cdot} \right] - \frac{3}{2} \frac{1}{m_\Phi^2} y_d^{i1i2} \\
& \quad \left(\overline{a_d}^{pr} \left(m_\Phi^2 c_\gamma^2 + 2 C_{H^2} s_\gamma^2 \right) + s_\gamma \tilde{\mu} c_\gamma \overline{y_d}^{pr} \left(2 C_{H^2} - m_\Phi^2 \right) \right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr} \right) LF_{3,1,-1} \left[m_q^{p\cdot}, m_d^{r\cdot} \right] + \\
& \quad 3 C_{H^2} c_\gamma y_d^{i1i2} \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr} \right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr} \right) LF_{3,1,0} \left[m_q^{p\cdot}, m_d^{r\cdot} \right] - \\
& \quad 9 C_{H^2} c_\gamma y_d^{i1i2} \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr} \right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr} \right) LF_{4,1,-1} \left[m_q^{p\cdot}, m_d^{r\cdot} \right] + \\
& \quad 6 C_{H^2} c_\gamma y_d^{i1i2} \left(c_\gamma \overline{a_d}^{pr} - s_\gamma \tilde{\mu} \overline{y_d}^{pr} \right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr} \right) LF_{5,1,-2} \left[m_q^{p\cdot}, m_d^{r\cdot} \right] + \\
& \quad 3 s_\gamma \frac{1}{m_\Phi^4} y_d^{i1i2} \left(C_{H^2} - m_\Phi^2 \right) \left(s_\gamma \overline{a_u}^{pr} - \tilde{\mu} c_\gamma \overline{y_u}^{pr} \right) \left(c_\gamma a_u^{pr} + s_\gamma \tilde{\mu} y_u^{pr} \right) LF_{1,1,0} \left[m_q^{p\cdot}, m_u^{r\cdot} \right] - \\
& \quad c_\gamma \overline{y_d}^{pr} y_d^{pi2} y_d^{i1r} LF_{1,1,0} \left[m_q^{p\cdot}, \tilde{\mu} \right] -$$